

PQC-Readiness Audit Report

Contract: /mnt/d/Crypto/audit_case_studies/contracts/CompoundComp.sol
SHA-256: fa2020d5b28cfc4a44e4383f41c5e35... **Lines:** 191 **Date:** 2026-08-20 **Tool:** pqc-audit v0.2.0

Executive Summary

Check	Result	Detail
Security Scan	WARN	0 critical, 1 high
PQC Readiness	FAIL	1 quantum-vulnerable primitives, 0 critical implementation advisories
Gas Optimization	25 opportunities	~14900 gas units
OVERALL	FAIL	

Phase 1: Structure

- **Solidity Version:** ^0.8.10
- **Contracts:** 1
- **Functions:** 18
- **Events:** 4

Phase 2: Gas Optimization (25 found)

#	Pattern	Impact	Fixable
1	Use custom errors instead of require with string message	Info	Manual
2	Use custom errors instead of require with string message	Info	Manual
3	Use custom errors instead of require with string message	Info	Manual
4	Use custom errors instead of require with string message	Info	Manual

#	Pattern	Impact	Fixable
5	Short-circuit evaluation	Info	Manual

Phase 3: Security Analysis (10 findings)

Severity	Count
Critical	0
High	1
Medium	6
Low	3

Security Score: 57.0/100

Top Findings

- **[MEDIUM]** Line 87 — Timestamp dependence
 - Code: `require(block.timestamp <= expiry, "Comp::delegateBySig: signature expired");`
 - Using `block.timestamp` (or `now`) in comparisons can be manipulated by miners.
- **[MEDIUM]** Line 85 — Missing zero address check
 - Code: `require(signatory != address(0), "Comp::delegateBySig: invalid signature");`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 125 — Missing zero address check
 - Code: `require(src != address(0), "Comp::_transferTokens: cannot transfer from the zero address");`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 126 — Missing zero address check
 - Code: `require(dst != address(0), "Comp::_transferTokens: cannot transfer to the zero address");`
 - Missing zero address check for critical operations.
- **[MEDIUM]** Line 140 — Missing zero address check
 - Code: `if (srcRep != address(0)) {`
 - Missing zero address check for critical operations.

Phase 4: Post-Quantum Cryptography Readiness

Line	Primitive	Risk	Migration Path
84	<code>ecrecover</code>	CRITICAL	Replace with ML-DSA-65 (Dilithium-3) via a FIPS 204 implemen...
22	<code>keccak256</code>	MEDIUM	Double hash output length or use SHA-3-512...

Quantum Risk Level: CRITICAL

NIST Status: FIPS 204 (ML-DSA) and FIPS 205 (SLH-DSA) finalized. FIPS 206 (FN-DSA/Falcon) selected for standardization; publication pending.

Implementation Vulnerability Scan (2024-2026 Research)

Vulnerability	Risk	Source
<i>None detected</i>	LOW	No known implementation vulnerabilities match this contract

Evidence Attestation

- **Source SHA-256:** fa2020d5b28cfcb4a44e4383f41c5e353bb6d75e733e0aebaf3b71eeb9332ed3
- **Report SHA-256:** eee5d18ea1cdc0a2f26b42c787c10914...

This report was generated mechanically. No LLM inference was used in the security or PQC assessments. PQC-Readiness Audit — pqc-audit v0.2.0